

Geometry
Unit 9
Lesson 7 Practice

Name _____
Date _____
Period _____

1. Austin lives by train tracks. He noticed a train goes by every afternoon. Most of the cars on the train are boxcars. A boxcar can be $60\frac{3}{4}$ feet long, $10\frac{2}{3}$ feet wide, and $11\frac{2}{3}$ feet tall. What is the maximum volume of a boxcar in the shape of a rectangular prism?
2. Austin counted 21 boxcars on one train Saturday. If every boxcar was full, what was the total volume of cargo on the train?
3. A ping pong ball has a diameter of 40 millimeters. What is the volume of a ping pong ball? Round to the nearest hundredth.
4. The minimum diameter of a golf ball is 42.67 millimeters. What is the minimum volume of a golf ball? Round to the nearest hundredth.
5. What is the difference in volume between a golf ball and a ping pong ball?

6. Albert went on a road trip. During the drive, he noticed that some water towers were spherical, and some were cylindrical. He learned that the first water tower he saw was spherical with a 20 foot diameter. What was the volume of the first water tower? Round to the nearest hundredth.
7. The second water tower was cylindrical with a 19 foot diameter and a 21 foot height. What was the volume of the second water tower? Round to the nearest hundredth.
8. Which water tower can contain more water?
9. An Olympic swimming pool is 50 meters long, 25 meters wide, and 3 meters deep. What is the volume of water in a full pool?
10. If the pool loses $\frac{1}{8}$ of its water depth during a free swim event, what is the new volume of water?